

KiCad overview

Beneath its singular surface, KiCad incorporates an elegant ensemble of the following standalone software tools. KiCad includes a project manager and comprises four main independent software tools:

Eeschema—the schematic capture editor.

Cvpcb—the footprint selector for components used in the circuit design.

Pcbnew—printed circuit board editor. It also has 3D view.

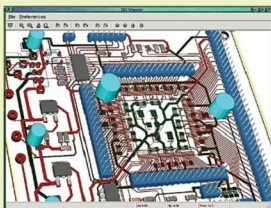
Gerbview—Gerber (photo plotter documents) file viewer.

Three utilities also included are:

Bitmap2Component. A component maker for logos (creates a schematic component or a footprint from a bitmap picture).

PcbCalculator. A calculator that is helpful to calculate components for regulators, track width versus current and transmission lines.

PIEditor. Page layout editor.



KiCad Pcbnew3D

Project

example.pro	project management file
example.sch	main schematic file
example.kicad_pcb	printed circuit board file
example.net	netlist file
example.xxx	various files created by the other utility programs
example.cache.lib	cache file of libraries used in the schematic (backup of the components used)

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each single schematic component with its footprint component. A very large component footprint library comes with KiCad.

Step 3: PCB layout. The Pcbnew board editor can handle up to 16 copper layers plus 12 technical layers (silk screen and solder mask) and allows you to layout the final PCB.

To successfully perform these steps,

KiCad comes with additional software tools. Two library component editors allow you to create or modify schematic components and footprint components. The 3D viewer allows you to render a final 3D model of your PCB.

For the generation of the necessary compliant files for manufacturing, your PCB (Gerber files for photo-plotters, drilling files and component location files), Pcbnew and Gerbview are

used. Postscript or PDF file generation is also possible.

What one can do with KiCad

KiCad is considered mature and can be used for the successful development and maintenance of complex electronic boards. KiCad does not present any board-size limitation and it can easily handle up to 16 copper layers and 12 technical layers. KiCad can also create all the files necessary for building printed boards like Gerber files for photo-plotters, drilling files, component location files and a whole lot more.

Being open source (GPL licenced), KiCad represents the ideal tool for projects oriented towards the creation of electronic hardware with an open-source flavour. It is an excellent tool for professionals and hobbyist, mainly due to the excessive cost of commercial software that are prohibitive for students and hobbyists.

With the help of the project man-

ager, one can start the creation of a project by launching Eeschema. It manages a direct and fast access to component documentation. Eeschema is an integrated package because it comprises all the functionalities such as library management, layout, drawing and control to name a few. Using multi-sheet diagrams, it allows hierarchical drawings—simple, complex and flat hierarchies. Needed for modern schematic capture, Eeschema provides some essential additional functions such as:

1. Design rules check (DRC) for the automatic control of incorrect connections, and the inputs of components left unconnected.
2. Export of the layout files to POSTSCRIPT or HPGL format.
3. Printing the layout files on a local printer.
4. BOM generation.
5. Net list generation for PCB layout or simulation software.

KiCad component libraries

KiCad comes with a large set of open source library components. A text-based format is used for both schematic and PCB components. This allows the direct editing of your library files with any text-based software.

Both Eeschema and Pcbnew have a library manager as well as a library component editor for modifying and creating components and footprint parts. You can create, edit, delete or exchange library items easily. Documentation files can be associated to components and footprints, and key words, allowing a fast search by function. Very large libraries, created over many years, are available for schematic components and footprints. Most of printed board modules (footprint) are available with their 3D shape model.

The other very exciting aspect of KiCad is that library components for both PCB and schematic are actually plain text files.

KiCad's principle of use

In order to manage a KiCad project, schematic files, PCB files, supplement-